

BOOST POWER BOARD - FABRICATION DRAWING

Release candidate BOOST_power_RC2, 2026-09-24 (rev6 values). Files: BOOST_power_RC2_gerbbers.zip. Units: mm.

FABRICATION NOTES

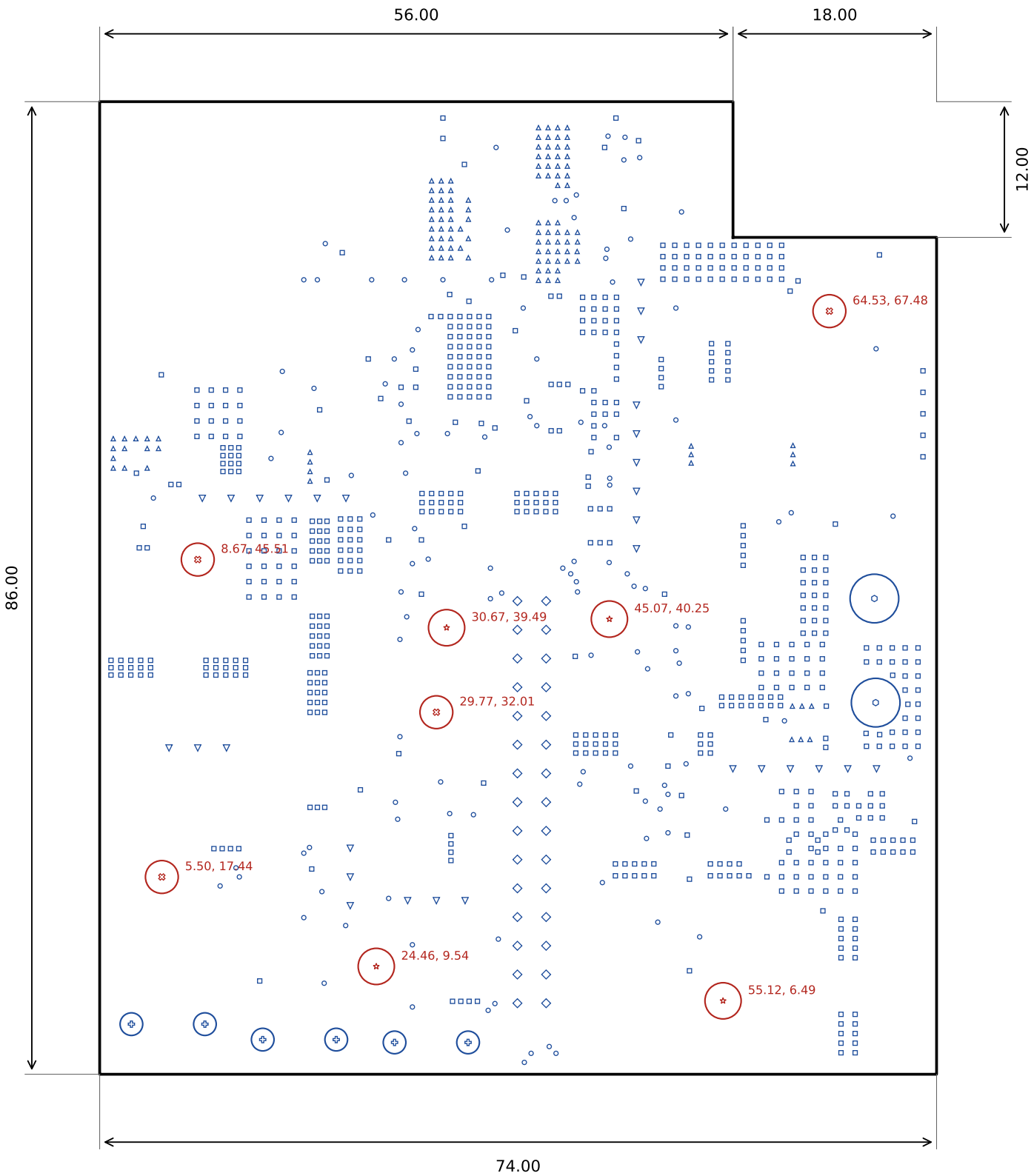
1. **ALL VIAS EPOXY FILLED AND COPPER CAPPED (resin plugged, plated over; IPC-4761 Type VII). 476 of the 868 vias sit in SMD pads. Via drills 0.30, 0.40, 0.50 mm (JLCPCB filled-via range 0.2-0.5 mm). Order option "Via Covering: Epoxy Filled & Capped".**
2. Material FR-4, Tg 155 C or better (JLCPCB "FR4 TG155", Nan Ya NP-155F). 8 copper layers. Finished thickness 1.6 mm +/-10 %.
3. Copper 1 oz (35 um) finished on all eight layers: outer 1 oz, inner 1 oz (order option "Inner copper weight 1 oz"; the fab default of 0.5 oz inner is NOT acceptable).
4. Stackup: the fab's standard 8-layer 1.6 mm build (no stackup specified), as in the table. No impedance control.
5. Surface finish ENIG. Solder mask green LPI both sides. Silkscreen white both sides.
6. Minimum track / space: 0.20 / 0.20 mm on every layer. Minimum drill 0.30 mm (vias); smallest plated pad hole 1.05 mm.
7. Minimum via annular ring 0.15 mm; minimum plated pad annular ring 0.20 mm. Via-to-via hole gap >= 0.25 mm; filled via to any pad hole >= 0.48 mm.
8. Plated hole tolerance +0.13 / -0.08 mm. J10 2x15 header holes are drilled 1.05 mm for Samtec 0.64 mm square posts.
9. Non-plated holes: 4 x 3.20 mm (standoffs), 4 x 2.90 mm (FET tab screws).
10. Outline on the Edge.Cuts (GM1) layer, tolerance +/-0.2 mm. Notched top-right corner.
11. Electrical test: 100 % flying probe; IPC-D-356 netlist BOOST_power_RC2.ipc356 supplied.
12. No order number or other marking to be added ("Remove Mark").

STACKUP (from the board file; copper + dielectric 1.654 mm, + mask)

F.Mask	solder mask		0.010
F.Cu	copper, 1 oz	signal / parts	0.035
	prepreg 2116 NP-155F, er 4.16		0.109
In1.Cu	copper, 1 oz	GND plane	0.030
	core NP-155F, er 4.23		0.250
In2.Cu	copper, 1 oz	signal	0.030
	prepreg 2 x 2116		0.218
In3.Cu	copper, 1 oz	signal	0.030
	core NP-155F		0.250
In4.Cu	copper, 1 oz	5 V plane	0.030
	prepreg 2 x 2116		0.218
In5.Cu	copper, 1 oz	signal	0.030
	core NP-155F		0.250
In6.Cu	copper, 1 oz	GND plane	0.030
	prepreg 2116		0.109
B.Cu	copper, 1 oz	signal / parts	0.035
B.Mask	solder mask		0.010

DRILL TABLE

sym	drill mm	type	count	used for
○	0.30	PTH	122	vias (filled, capped)
□	0.40	PTH	628	vias (filled, capped)
△	0.50	PTH	118	vias (filled, capped)
◇	1.05	PTH	30	2x15 socket (J10)
▽	1.10	PTH	30	TO-220 pins M1-M10
⊕	2.00	PTH	6	LED wire pads J3-J8
⊗	2.90	NPTH	4	FET tab screws
★	3.20	NPTH	4	standoffs
◯	4.30	PTH	2	battery lugs J1/J2



Datum: lower-left corner of the outline (0, 0). Dimensions in mm, outline tolerance +/-0.2 mm.
Drawn from the Excellon files: every drilled hole is shown with its symbol; holes >= 2 mm also to scale.